

DEF CON 25 Conference - Alvaro Muñoz, Alexandr Mirosh - Friday the 13th JSON attacks

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Description

2016 was the year of Java deserialization apocalypse. Although Java Deserialization attacks were known for years, the publication of the Apache Commons Collection Remote Code Execution (RCE from now on) gadget finally brought this forgotten vulnerability to the spotlight and motivated the community to start finding and fixing these issues.

One of the most suggested solutions for avoiding Java deserialization issues was to move away from Java Deserialization altogether and use safer formats such as JSON. In this talk, we will analyze the most popular JSON parsers in both .NET and Java for potential RCE vectors.

We will demonstrate that RCE is also possible in these libraries and present details about the ones that are vulnerable to RCE by default. We will also discuss common configurations that make other libraries vulnerable.

Transcript

Retrieved with yt-dlp in a container, which is the only route YouTube still answers. Expect transcription errors in type and method names.

hi def con I'm Alberto pond tester this is Alex as we said we both work for the HP fortify team but we have a lot of things to cover today so enough about us so yeah last year was the year of javadi's realization apocalypse as we like to call it like literally dozens or even hundreds of new series were reported around Java this realization and around those TVs like thousands of vulnerable applications in appliances it was like a bad problem so this was a very well known vector since at least 2011 probably it was not so the community was not paying I really like attention because there were no a really good cut yet I really a good remote code execution gadget until 2015 when the Apache Commons collection gadget was published so this was the same problem with dotnet the theory I'll attack was published in 2012 for both binary format there and the net data contract CD Eliza but there was no remote code execution gadiot publish until now so at that time the solution was like okay I stopped using Java the serialization all together and developers were like well yeah I would like to do so but I need to serialize my object store them into a database or maybe I don't know send them across Hawaii or send them in a socket or whatever so I need to serialize my object and they were and the security guys were like okay then use a secure json or xml parser or library instead so our goal with this talk is twofold first of all we want to verify if those json libraries are any better than Java digitalisation or donate the serialization and then raise some awareness around dotnet visualization staff always have with Java it took like four four or five years for the community to start looking into you know vulnerable endpoints and so on and we don't want to the history to repeat itself so we will be presenting new remote code execution gadget to attack this VM serializers so the first section will focus on on Jason we will present some of the libraries that we found to be vulnerable and gadgets to attack them and then a demon a web application framework leading to arbitrary code execution and we will focus on the donuts realizers completely a different subject and then we will try to generalize attack and we will pry Vince and Amos as well so let's jump into the Jason part so yes to set the expectations here we are not talking about Jason when sending simple data our simple JavaScript objects between a Java JavaScript front-end and a controller like you know for example a supreme controller we are talking about replacing Java or.net serialization with a JSON which means that you need to serialize Java or.net object which means that you need to support some of the object-oriented programming features like how I'm going to deal with the Java language yet I'm going to deal with a system object in dotnet I am going to deal with generic parametrize types polymorphism interfaces and so on so we will be abusing the the features that these libraries include to support these auto P stuff very quick recap on how Java digitization or how the attackers were able to get a bit ready code execution on jered visualization basically they were able to provide untrusted data to the base

réaliser in this case the object input stream in in Java and this trusted data contained some type some class names that were instantiated by the visualizer and then some visualization callbacks like read object or with read result where invoke and then the attackers were able to start a catcher chain that is nothing else that connect in different classes together assembling like a Katia chains leading to arbitrary code execution but a starting from this read object or read resolve a DC allocation callbacks however Jason does not invoke any of these DC allocation callback normally or at least some of the most of the libraries so we need to find some some other ways to start our gadget chain leading to this code execution so most of the JSON libraries will work in will we'll try to reconstruct the object in any of these ways so basically they will invoke the default constructor in order to instantiate the object and allocate the memory and then use either reflection or setters in in order to populate the fields and properties of the of the instance some of them will invoke some special constructors or deserialization callbacks or type converters we don't have time in this talk to cover all of them but these are also vectors that lead to arbitrary code execution we have we have them covered in our white paper that has will be published in the Deaf commedia server and then most of these libraries also invoke some common methods like for example house code or ake was normally get invoke when DC arise in hash tables to a string normally get involved when raising exceptions finale's is always invoked by the garbage collector which when claiming the memory for the objects but by far the most common one is the setters setters have normally invoked by most if not all of the libraries so if we are able to find arbitrary code execution gadget chains starting with setters we will be able to use them to attack these libraries let's have a look on a few setter gadget in dotnet all of them can lead to arbitrary code execution some of them has own requirements and limitations but we believe that it's not very difficult to pick up proper one for specific case so our first gadget is the set of path property in assembly installer class it allows code execution during library loading from path control bonanza by an attacker it learnt is no additional requirements if assembly this payload is on local machine but in case if remote resources dotnet framework may have some additional security checks the next two guarded use uses XML parser we will show a bit later how it can be used for arbitrary method invocation so a setter of property inspector font and color data in workflow designer type requires single threaded apartment set it's quite strong require a requirement but if your target has such configuration you will get remote code execution the next gadget is source setter of research dictionary type it has a few requirements to JSON unmarshal it should call setters for types that implement I dictionary interface and often in such cases and marshals are just populating key value pairs also it should be able to reconstruct system override object this tab type doesn't have default constructor but often my own marshals can do this and our final gadget is object data provided type it's quite flexible and allows a lot of ways to for attack as a result we were able to use it in almost all our and marshals and formatters so let's have a look on call diagram of this guard etcetera will call refresh method it will invoke begin query it will call query worker and finally in invoke method on instance we can see the line that will call our arbitrary method here we can see example of JSON payload that will pop-up calculator for unsafe configuration of JSON net parcel actually this gadget allows the next main attack vectors we can call known default constructor with own arguments or can invoke public

method on unmarshal eight object or we can call any public method including static ones with our own parameters Java has own a setter gadget as well after all our last year research in JDI injection attack we have found a few setters with jin-gi lookup caused by the way oracle recently disabled ARIMA and Corbeau vectors in the fall configuration of java but aldub vector still works so our gadget set session factory genie name in statistic service we already mentioned about this in our last year talk at blackhat 2016 very similar to it it's to string gadget from remote client user transaction clasp and finally set of the community graduate from already known class gdb 0 set in po let's have a look on the it this class is from GRE libraries so doesn't trougthon doesn't require any external dependencies it's attribute called connect function and here we can see the typical in the initial context lookup method with our value from data source name properties so we will get remote code execution apart from my shotguns that allows code execution by themselves we have a couple other interesting gadget some of them can be used as building blocks for gadgets chain like Bendix or said that a member in dotnet or string template to string in Java can be used for arbitrary got Gator calls other can trigger known RC attacks for example in a XML in XML data document or set of data view setting collections in in that the view manager can be used for XML external entities attack in some version of .net framework now we will back to Alvaro he will show where we can use this all gadgets so we analyze like multiple Java and .net the additional libraries and we found that most of them were burnable toured with radical execution some of them in their default configuration and some of them developers need to enable or configure them in a special way so we came up with these simple high-level requirements for our library to be Berner able to this attack basically the attacker needs to be able to control the type or the class that is going to be instantiated in the server that normally means that in the JSON that you are intercepting for example there will be an attribute called something like underscore type or dollar type class type object class name that contains a value that looks like a doughnut or a Java class name then the library will instantiate that type and then the second requirement is that the attacker or well in this case the library needs to invoke some methods on this reconstructed object and finally the third requirement is that the attacker needs to be able to assembly remote code execution graduate chain starting from those methods that are invoked by the by the library as I said before setters are normally book in most of their libraries so they cut it that I Alex as presented are normally can be normally used for any library so we categorize these libraries according to the likelihood of them being vulnerable and for that we choose two different factors the first one is whether the library includes the type discriminator that is nothing else that the class name in the sterilized data like we're seeing here for example in JSON dotnet and so that can be done like by default by they come by the library or developers may need to enable a configuration setting and then the second factor is how this library prevent malicious types from being instantiated and loaded so we found that some of these libraries perform a post DC allocation cast which offers no security at all because basically by the time that you get at the cast exception the payload has already been executed and you're already been compromised and the second type control is what we call the inspection of the expected type object graph which sounds like a little bit complex but it's nothing else that the library will take the expected type so for example a user the library may be expecting a user or a

cart or whatever and then analyzing the object graph and checking if what the user is sending in the indigestion data is a signal to the types defined in the object graph so some libraries go further and also build a whitelist at construction time and apply this list at run time during the disk realization we found that these these libraries the libraries performing this type control are also vulnerable if the attacker can control the expected type that is more common than you may think and we will see an example an example later and also if the attacker can find an injection point and entry point in the object graph so for example as we can see here in this sample object graph here all the properties in red may be good examples of entry points like for example a system object property a non-generic collection property like hash table array list and so on so again we don't have time to cover how to find entry points but this is also covered in the in the white paper so this is the list of the additional libraries that we analyzed there once in red vulnerable by default so fast JSON a sweet JSON JSON i/o on PlayStation should never be used with untrusted data basically they include the type information by default and they perform no type control at all then they invoke setup so we can use our setters to get a arbitrary code execution the ones in yellow and orange depends on the developers and how they are using the library so most of them do not include the type information by default in the JSON data but there is this configuration setting that can be enabled to do exactly that and then they perform this inspection of your graph so you need to find an entry point you may think that a finding an application that satisfies both requirements is difficult but it's kind of the opposite because if developers has a property or example of system object they will be forced to enable this configuration setting to include the type information and then make this class or this type set a high school so well gisun is the only one in green we found that is very difficult to make it vulnerable you can also implement a type a lab there and make it vulnerable but I mean you need to have to do it like in purpose and in addition it does not invoke any set there it just use reflection so you need to use or to fine cut yet starting for example with a finalize method which is more difficult than than the setter one so let's see some of these examples for example first jason is one of the ones in red should never be used with untrusted data and basically yeah it includes the type information by default it does not offer any that I put type protection or type control and then it will invoke the setter so we will be able to use for example our object data provided cut yet to get arbitrary code execution like we did in this calico CMS that is a dotnet content management system then we have a JavaScript serialize er that is one of the dotnet native libraries in the .net framework by default it does not include the type information but developers can do that by passing this simple type resolver or any other type resolver to the constructor of the JavaScript serialize ur and then DC allies in untrusted data so by the way all these examples are from github so this is real code basically this JavaScript serial I said does not offer any type control it performs a post DC allocation cast operation so it it's moon revolt by default if developers are configuring it with a type of solving then we have the data contract decency realized that is probably one of the most secure ones but well it performs this sustainability check that we talked before but also this creation of a whitelist at construction time that is then applied at runtime time so with that the only way a develop an attacker can get arbitrary code execution on this library is if they can control the expected type again this is from github it's an example of how the type can be

controlled by the attacker in this case is coming from a cookie we will see an example in DNN that is one of the most popular CMS for donut where they are doing exactly this and then we have JSON dotnet that is probably the most popular libraries even recommended by Microsoft over their own native libraries again by default does not include type information but if developers enable this table name handling setting then they will include this information this is another example from github as you can see because it contains this class are called message because it contains body property which is a system object type then developers are forced as I said before to enable these configuration settings to include the type information and then the library will invoke setters serializable constructor callbacks and even some own error custom callbacks so basically the attacker has like a very large gatita space to get arbitrary code execution so let's see an example a demo this is breeze that is a web application framework which basically what it's not really a web application framework is more a data management framework so it offers a REST API for both JavaScript and dotnet clients to connect to the API and perform like entity management operations like having a new record the latina record a bear in a record and so on so we found that this framework was used in JSON net which is like as I said a the most popular JSON library for .net and is configured to include the type information when satirizing the system object properties so at some point this code here will digitalize untrusted data coming from the request and then as you can see here basically these safe options type here is the expected type so I said that we have to do this expected type object graph inspection which is quite simple in this case because the safe option type only contains two properties the second one is name tag and it's a system object property so we can this is our injection point our wearing three point two place the the payload so with that let's see an example this is carbons that is one of the sample application that comes from with the with a framework for developers tool to learn the framework and basically in this example we can just up new car models or rename the models and so on so if we change the name of this model and we intercept the request with burp we will see that well we have all the expected data like for example the new model name is appearing in the indication data that is being submitted to the REST API and then we have this empty safe options dictionary if we send this request to the server we will see that there is in the response that is this dollar type attribute which is our indicator that this is using JSON dotnet and that is including type information so that's those are the ingredients for our arbitrary code execution we will be using our object data provider gadget basically to invoke the process dot start and pop up a calculator when the framework process this request so we will just replace this empty dictionary with the payload and then let's check that in the victim server there is no calculators running and now as soon as we submit the request they will get this realized and we will get arbitrary code execution [Applause] [Music] so thank you so this is a popular rather popular a framework in donate and a big set out to the developers because they fixed the issue in just one day also a big shout out to Morris Whistler that is a Java security researcher May 22 he published this Java Marshall security paper which basically similar to our research serves the same premises and conclusions will he overlaps with our research in the Jackson and Jason are your libraries although for the day someone he found like a completely different vector and he also overlaps with us in our JDBC roasted in poll class or cat yet but this was like a kind of like

obvious choice because it was previously used in the Java visualization work also he found other interesting gadgets for Java in third-party libraries like a spring so if you are interested in a more java focused version of this talk go and check his his work so now we will change gears and focusing that naturalizers let's back to the dotnet world potential security problems with binary format and net data contract serializer were known for the year for example the great work of James for show about the weaknesses and main attacks in in this main dotnet formatters was presented at blackhat 2012 so five years ago unfortunately we could not find good available remote code execution gutter chain there was one published by Florian gold here but it uses memory corruption so it's not very easy to build Universal exploit for different version of Windows but we were sure that there should be a lot of ways to get code execution during dotnet digitalization so we spent some our time and have found one that can be used in binary format and some other dotnet for matters but after our work was ready and accepted by blackhat the same James for show a couple remote code execution gadget as part of his own research by the way not connected with our topic anyway you can find the tell us about this gadget on his post on Google project zero block here we will focus on our own we will use PS object type it is part of PowerShell libraries so should be available in almost all Windows machine before we go further there are couple remarks about this gadget in pivara version 1 and this type is not serializable so is not suitable for attack but all modern version of Windows starting from Windows 8 and Windows Server 2012 are shipped with newer and - honorable version the next remark we reported this issue to Microsoft and two weeks ago and they released a fixed so if you are not if you are not ignoring their updates you should be safe so PS object for the civilization it use custom the sterilizer and in case of TI m instance it will call CI rag'd right see I am instance property method here we can see that attacker is able to specify own type as element type of array and the sterilizer will try to reconstruct object of this type to find proper way to do this it will use figure conversion matter it's quite interesting method for an attacker as there a lot of options for attack we highlighted only the most obvious one so we can call non default constructor with one argument that we can control its value also we can invoke setters on public properties so we can use our guards earlier gadgets and also we can use static parts method of arbitrary type let's try to use this one sorry as we said earlier some redraw parts can use you can be used for arbitrary method invocation here we can see examples of payload that call proces start with our argument here we can notice namespace assembly type method name and finally calc as an argument along with mentioned binary in data contract serializer formatters dotnet offers a lot of others unfortunately we don't have enough time for the teletrac view of each of them so we will briefly cover them in our today talk and if you're interested in deeper analysis we can offer out white paper as good source so we can group them into big group format they are vulnerable interval configuration like binary format or table format no data contract serializer also we can add here for matters that internally use them like object state formatter lost format binary message format they should not be used with untrusted data or you have to proper configure them to limit available types for example use restricted type resolve with white list to the types other group for matters that are safe in default configuration like XML serialize data contract serializer data contract JSON serializer but if you are using quick data contract resolver by the way we

have met such examples even on official Microsoft documentation or attacker is able to control control expected type you will have very serious security problem we will show a bit later that in this case code execution is real even for for the most limited for matters for example XML sterilizer now switch back to Alvaro he will show our next demo so because Hadi most probably worth a thousand slides we will show you how to get arbitrary code execution on some dotnet framework in this case Nancy that is a web application framework that is very similar to the Ruby's Sinatra framework but for the net and well basically they care about security which is good and they implemented CSR protection the only thing is that by protecting against EF they opened the doors for remote code execution but so instead of putting this unique token into the cookie like most of the frameworks do they created like a type causes of instance where they put this token and then they serialize this instance of cecil token and then they basically for encode the payload until the payload they set the cookie into the cookie anyway if you find this a a a d magic number in any of your pen test then you will get you will be able to get arbitrary code execution so let's see that in action in one application so this is an application we build you seen Nancy and we were able to get arbitrary code execution pre authentication by basically replacing the cookie so this is the cookie that the framework injects the ends is our cookie as you can see it contains this magic number and then this base64 encoded block if we check the page they also include the same value as a hidden filled in in the form so they are doing like a double submission which is good but since they are using binary format of civilization then it's not that good and we we see realize how where PS object cut yet in order to pop up a calculator with a calculator payload and then we are just basically replacing the cookie with our payload you can use our payload or any of James for some published payload and now it will go back to the server there are no calculators running and as soon as we submit the form we saw what will we will get an exception but we will get our payload executed thank you so we just wanted to highlight that this is not really a problem in JSON or XML parsers or I don't know Java visualization or the binary formatter for dotnet this is a preliminary license themselves because all of them need to reconstruct the objects during the digitization process and that means instantiating types that normally can be controlled by attackers and then invoking methods so as we said before this simple three high-level requirements like the attacker being able to control the type then the some methods being called on this reconstructed type and then a large enough cut it the space for the attacker to find arbitrary code execution gutter chain will normally apply to any serialize here on any language as we said before most of the times the setters will be invoke so they catch it that we presented here today the object data provider cut yet and the JDBC rows settle in pull gadgets will normal will normally be good cut just to attack any other libraries so we found like many other libraries with invulnerable or being as susceptible to arbitrary code execution but we don't have space in the slide or time to cover all of them so these are good examples of them for example FSP clear or subsidy eliezer are the nets realizers that work not just in the dotnet framework but also in dotnet core Silverlight are windows mobile phone for example so developers may choose them instead of the native one because they are suitable for their needs and we'll all of them include type information by default they invoke setters and they perform either no type control at all or just the object graph inject inspection so

attacker may need to find these entry points also we have wire that is now known as Hyperion that is the sensor for a cathode net if you are familiar with the Scala or Java version of the libraries framework for concurrency based on an actor so you have like different actors exchanging messages and these messages were as realized with JSON dotnet using this type name handling set in equals 2 also it was vulnerable but then they replaced this JSON dotnet realizer with hyper that is their custom sterilizer and will this Hyperion includes type discriminators invoke setters and then perform note type control so again is another vulnerable library also a beware of rolling your own format for example this is Nancy framework that we talked about before they moved away from binary format there because well it was not included in the first version of .net core and they want their framework to be compatible with don that core so they moved away from binary format al and then they created their own JSON library and as you can see here it includes a type discriminator that is called type object in this case and then it will invoke this setter so we were also able to get arbitrary code execution on the pre-release to the text branch of this framework we also found a very interesting CMS as one of the most popular CMS for for the native if not the most that is called DotNetNuke and they use xml serialize at betray types since xml satellizer need to know the type ahead of time then they created this wrapper around xml settle I said where they would in a cookie in a web cookie they were setting the type that was going to be used for the donut XML sterilizer and then the payload for the XML serialization so this was like a challenge for us because XML satellizer as Alex said is the most limited satellite internet it basically doesn't allow you to do many things like serialize in a system type property or set a lysine at type for example with interface members so if we were able to get arbitrary code execution and XML settle I said we were like confident that we were going to be able to do that for any other libraries so we tried with our object data provider gadget and so far so good it was like XML satellizer friendly no problem so far now if we try to use that card yet to invoke process dot start then we had a problem because the process type contains interface members so xml's eliezer doesn't like it that problem was easily solved because you can choose cards from anywhere else like for example the sam'l reader load method the object state formatter this realized method or you can even for God yet in your target class and class path or type so for example since we are attacking dotnetnuke we search for gadgets in DotNetNuke and we found that they have this file system utils type which is very handy for hackers because they contain methods to deploy web cells and methods to read any arbitrary files into the setp response so very convenient for us so now the next problem was that XML say that Isis builds a wide list of allowed types at construction type and then it applies the whitelist at runtime runtime time so that was a problem because we need to send an instance for example of our sam'l reader or the file system utility types as the payload and these types are not going to be included in the whitelist so we needed to find some way to fool XML satellizer to include our arbitrary types into the whitelist and we did that basically by using a parameterised type as the expected type as you can see down there and then we use the first parametrized type to place our payload type like filesystem utyos' and then the second type to put our object data provider type so with that we were able to get arbitrary code execution pre authentication on DotNetNuke at first these guys from dotnetnuke assign it like a low severity which was like weird because it's pre

authentication RC and they said that this severity was assigned because the attacker needs to be able to understand how the cookie works and we were like ok do you know the other framework is open source right so anyway they changed that to critical at the end so this is the source code for the relevant part at some point it reads for example when you visit the 404 error page it executes this code and as you can see here I will read in the cookie with dice name dnm personalization and then pass in this XML cookie into this the serialize hash table XML method if we check this method is basically extracting the type from the XML cookie like here and then using this type name to construct the XM satellizer so the attacker can control the expected type as we said before now is deserializing the rest of the cookie so this is what DNA em personalization cookie looks like in a real tienen application as you can see here it contains the type and then the inner blue box is basically the payload for the XML ser Alliser and this is what our payload looks like so the expected type contains this parameterized type which contains our runtime payload types and then the inner blue box is our payload for the object data provider so in this case instead of popping up a calculator we are using the file system utils class in DotNetNuke to basically download our website and deploy it into the IAS root folder so let's see that in action so this is at the out-of-the-box installation of DNN if you run it visit a non-existent page you will get your 404 error page sorry this page you are looking for it's not here and if we send that burp well the request does not contain the dnm personalization cookie but by reading the code we know that is processing this cookie so we will just include it manually like DNN personalization and then we will basically paste the XML payload that I saw you before which as I said is using the pool file method in the filesystem URIs class that is provided by the DNN to deploy the web show so let's paste this into the request format is a little bit and now in the victim server we will check that well we don't have this cell.aspx website that we are downloading so now as soon as we execute the request the cell.aspx will appear as you can see there and well be able to basically interact with the system basically by visiting the webpage and install whatever we wanted thank you so without just to wrap up just remember that this is not a problem in JSON or XML or I don't know any of the civilizer so try to avoid this LIC untrusted data we saw this is also problem in Ruby Python PHP so do not decision trusted data and if you have to get your library evaluated by some security guys try to avoid libraries that performs no type control so they are just doing the post DC allocation cast try to avoid libraries that include type discriminator is not like it's not a sufficient requirement for arbitrary code execution but it's a good indicator that you will be able to get up with a record execution and also don't allow developers to let users control expected type like we saw in DNN and last but not least try to not roll your own format because you can just it up so with that thank you very much and if you have any questions Karina's [Applause] [Music]